

Blending Synthetic Anomalies for Anomaly Detection in Human X-ray imaging

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Winter 2024

Introduction

- Manual X-ray analysis is prone to human error such as fatigue and subjectivity.
- In the context of security X-ray applications, Computer Vision methods are limited by the lack of positive data containing security threats.
- Current methods of introducing synthetic anomalies are insufficient for security X-ray imaging because they rely on existing parts of the image [1].
- Our Contributions:
 - Exploring the effects of pretraining on synthetic anomalies for Anomaly Detection methods using the Student-Teacher architecture.
 - A novel method of introducing synthetic anomalies to X-ray images (Figure 1).

Abstract

Anomaly Detection in Computer Vision has proven to be capable in using normal data to detect abnormal data. In this study, we explore how using synthetic anomalies affects the performance of Anomaly Detection architectures that rely on pretraining a teacher module. We further introduce a novel method of creating synthetic anomalies in the context of human X-ray imaging.

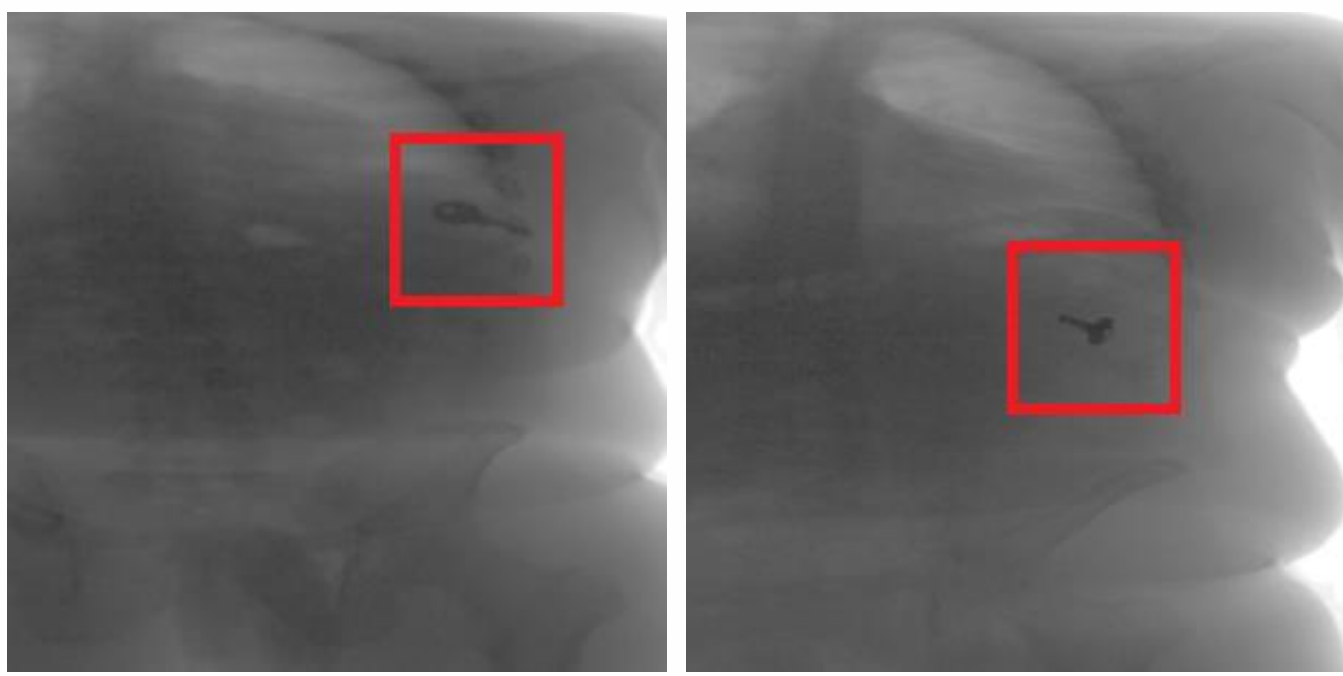


Figure 1. Comparison between a real anomaly (left), and a synthetic anomaly generated by our methodology (right). Highlighted in the red box on the left is a key found in the X-ray scan of a human body, while the one on the right is an anomaly image blended into the X-ray to imitate a real key.

Materials and Methods

- Models used:
 - Efficient AD [2] (Figure 2)
 - PatchCore [3] (Figure 3)
- Synthetic data with a difficulty factor to adjust anomaly visibility (Figures 4a, 4b).
- Real anomalies annotated to measure performance on real anomalies.
 - Anomaly detection measured by AU-ROC on the image level.
 - Anomaly localisation measured through AU-PRO and F1-Score (PatchCore).

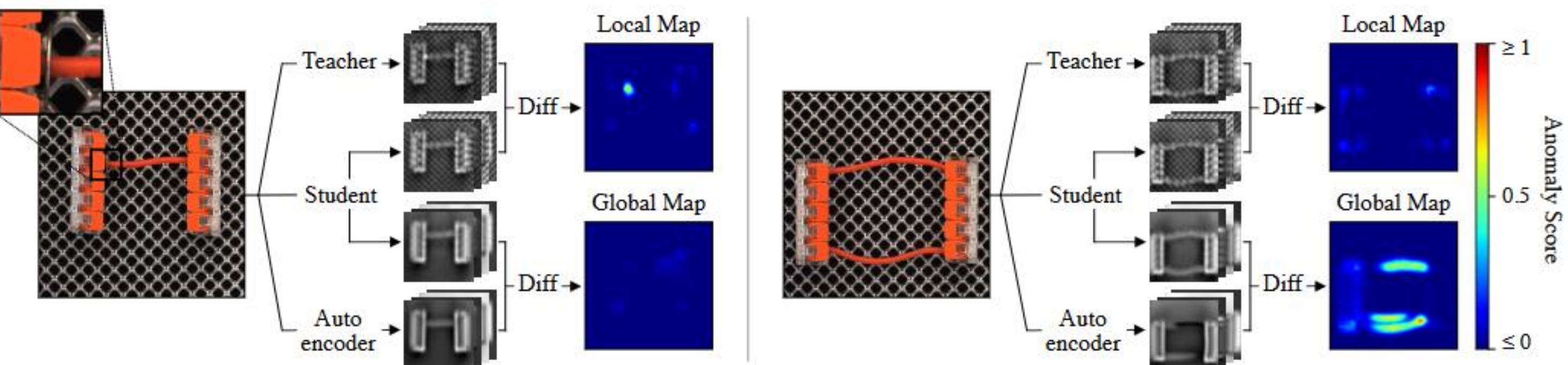


Figure 2. Architecture of Efficient AD [2]. It incorporates the Student-Teacher architecture to detect structural anomalies, while introducing an autoencoder module to detect logical anomalies.

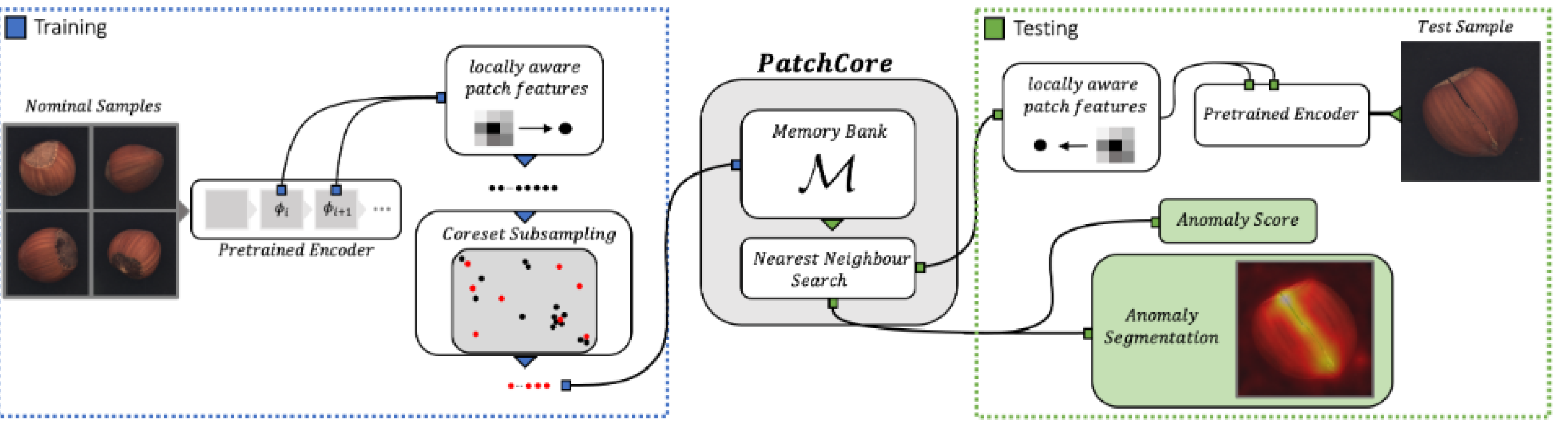


Figure 3. Architecture of PatchCore [3]. It breaks samples down into a memory bank of neighbourhood-aware patch-level features and detects anomalies by scoring each patch-feature.

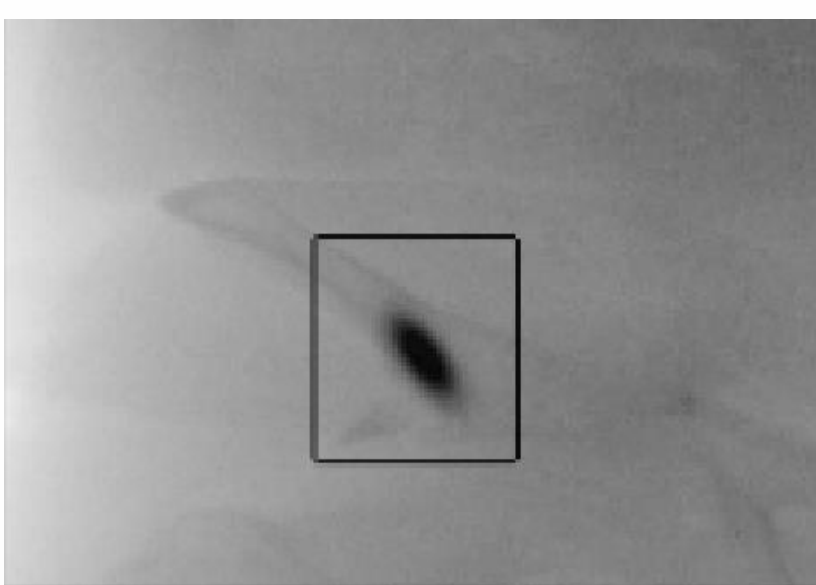


Figure 4a. Synthetic anomaly generated at difficulty factor 0.

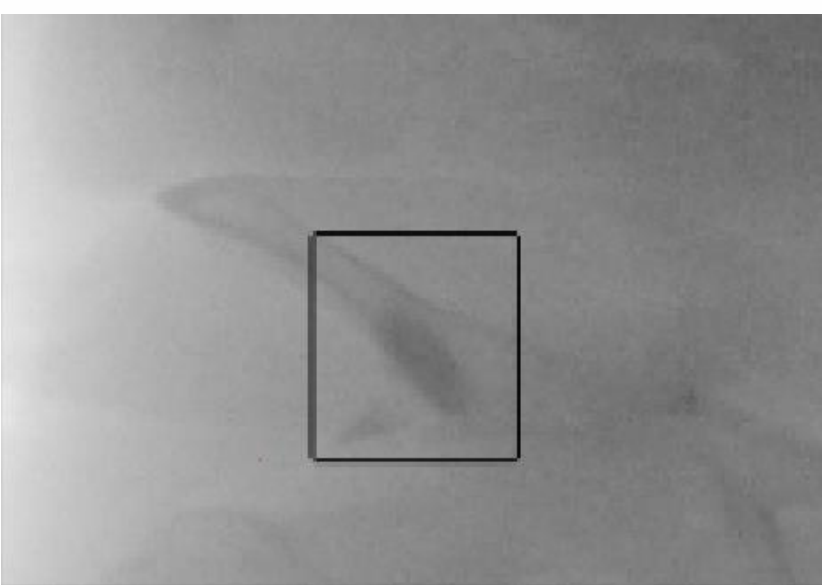


Figure 4b. Synthetic anomaly generated at difficulty factor 0.55.

Acknowledgements

- This project is a collaboration between the Computer Vision & Data Science professorship and ODSecurity B.V.

Experiments and Results

- Results of Efficient AD model against real anomalies when pretrained on datasets of incremental difficulty factors.

Dataset	EfficientAD-S		EfficientAD-M	
	AU-ROC _{Image}	AU-PRO	AU-ROC _{Image}	AU-PRO
df .0	0.53	0.16	0.63	0.17
df .15	0.55	0.15	0.58	0.16
df .33	0.51	0.23	0.53	0.13
df .45	0.65	0.15	0.59	0.16
df .55	0.57	0.15	0.57	0.16
df .70	0.61	0.15	0.58	0.17

Table 1. Results of testing both Efficient AD variants on real anomalies when trained on synthetic datasets of incremental difficulty factors (df). The results show that there is no significant effect in the performance towards real anomalies when pretraining is done on the synthetic data.

- Both versions of Efficient AD (small and medium) were used for training (Table 1).
- Teacher trained by distilling WideResNet-101 into Efficient AD with our synthetic data.
- Datasets are identical in terms of anomaly type and position in each image, except for the difficulty factor applied.
- PatchCore used to explore decline of performance on synthetic anomalies (Table 2).

Dataset	PatchCore	
	AU-ROC _{Image}	F1-Score _{pixel}
df .0	0.79	0.22
df .15	0.76	0.19
df .33	0.69	0.13
df .45	0.63	0.09
df .55	0.57	0.09
df .70	0.51	0.01
real	0.74	0.27

Table 2. Results of PatchCore on synthetic anomalies and real anomalies. PatchCore demonstrates similar performance between synthetic anomalies at difficulty factors (df) between 0.15 and 0.33, and real anomalies.

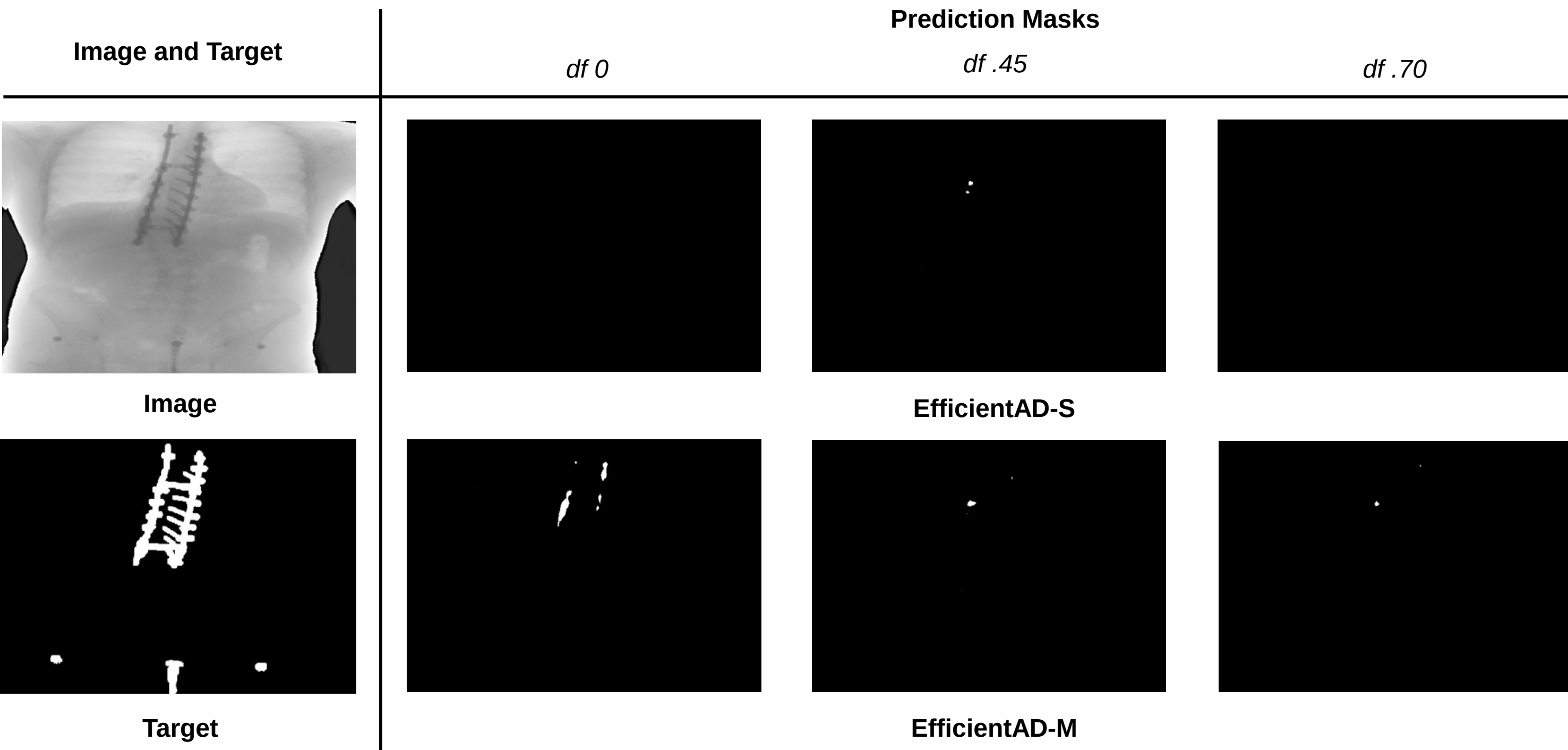


Figure 5. Results of EfficientAD-S and EfficientAD-M models on the same image, when pretraining is done on datasets of different difficulty factors (df). The results of EfficientAD-M across the difficulty factors may mean that pretraining on synthetic anomalies of similar intensity values to the type of real anomalies present improves performance, but the results of EfficientAD-S contrasts this hypothesis.

Conclusions

- Using synthetic anomalies during pretraining allows anomaly detection models to detect anomalies in real data.
- The use of different difficulty factors to introduce more difficult-to-detect anomalies does not have a significant effect on the ability of the model to detect anomalies (Figure 5).
- Our method of creating synthetic anomalies has the potential to represent real anomalies for anomaly detection.

References

- [1] Hannah M. Schluter, Jeremy Tan, Benjamin Hou, and Bernhard Kainz. Natural synthetic anomalies for self-supervised anomaly detection and localization, 2022
- [2] Kilian Batzner, Lars Heckler, and Rebecca Konig. Efficientad: Accurate visual anomaly detection at millisecond-level latencies, 2023.
- [3] Karsten Roth, Latha Pemula, Joaquin Zepeda, Bernhard Scholkopf, Thomas Brox, and Peter Gehler. Towards total recall in industrial anomaly detection, 2022.



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